Today: Poisson process continued:

- superposition of independent Poisson point processes
- thinning of a Poisson point process

these are process analogues of distributional identity

$$X_1 + \cdots + X_n \overset{d}{=} \text{Poisson}(\lambda_1 + \cdots + \lambda_n) \quad (*)$$

if $X_1, \ldots, X_n$ independent and $X_i \overset{d}{=} \text{Poisson}(\lambda_i)$

- examples

## Superposition

viewed as random set of points

if $X_1, \dots, X_n$ are independent Poisson point processes with intensities $\lambda_1, \dots, \lambda_n$

then $X_1 \cup \dots \cup X_n$ is Poisson p.p. w/ intensity $\lambda_1 + \dots + \lambda_n$.

two ways to prove:

① showing (*) directly by convoluting distributions:

for $n=2$ & Poisson random variable $X_1, X_2$

$$P(X_1 + X_2 = k) = \sum_{j=0}^{k} P(X_1 = j) P(X_2 = k-j)$$

then plug in
Poisson distribution:

$$e^{-\lambda_1} \frac{\lambda_1^j}{j!} \, e^{-\lambda_2} \frac{\lambda_2^{k-j}}{(k-j)!}$$

rearrange the sum into

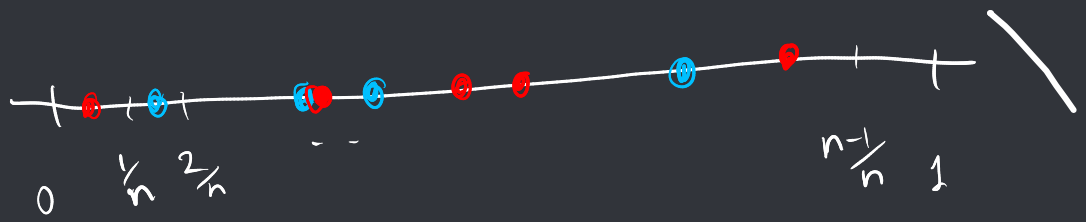
$$e^{-(\lambda_1+\lambda_2)} \frac{(\lambda_1+\lambda_2)^k}{k!}$$

using $(\lambda_1+\lambda_2)^k = \sum_{j=0}^{k} \lambda_1^j \lambda_2^{k-j} \binom{k}{j}$

② think of construction by subdivision & Bernoulli trials: ($n=2$ case)

$\lambda_1$

$\lambda_2$




$P$ ( blue point in $k^{th}$ subinterval ) $= \lambda_1/n$

( red ... ) $= \lambda_2/n$

$P$ ( neither ) $= \left(1 - \lambda_1/n\right)\left(1 - \lambda_2/n\right)$

$= 1 - \dfrac{(\lambda_1+\lambda_2)}{n} + \lambda_1\lambda_2/n^2$

$$P(\text{red OR blue} \quad \cdots \quad) = \frac{(\lambda_1 + \lambda_2)}{n} + O\left(\frac{1}{n^2}\right)$$

corresponds to PPP w/ intensity $\lambda_1 + \lambda_2$

This construction also gives intuition for

"independent minimum & minimizer"

property of independent exponentials:

Starting from $t=0$, time of $1^{st}$ (red OR blue) point:

$$= \frac{1}{n} \text{geo}\left(1 - \left(1 - \frac{\lambda_1}{n}\right)\left(1 - \frac{\lambda_2}{n}\right)\right)$$

$$\approx \frac{1}{n} \text{geo}\left(\frac{\lambda_1 + \lambda_2}{n}\right) \xrightarrow{n \to \infty} \exp(\lambda_1 + \lambda_2)$$

Conditioned on first point in $k^{th}$ subinterval,

$$P(\text{first point is blue}) = \frac{P(\text{blue point in } k^{th} \text{ subinterval})}{P(\text{blue or red point " " "})}.$$

$$\approx \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

# Thinning

start with Ppp w/ intensity $\lambda$
& independently colour each point
blue w.p. $p$ & red otherwise.

Then (blue points) & (red points) are
independent Ppps w/ intensities $\lambda p$, $\lambda(1-p)$.

Why independent? Because events with
small probability are "almost independent"!

if $P(A_1) = a_1$, $P(A_2) = a_2$

then $P(A_1 \cap A_2) \approx a_1 a_2$

(exercise)

In context of
discrete construction:

in each subinterval

$$P(\text{blue point}) = \lambda/n \cdot p$$

$$P(\text{red point}) = \lambda/n (1-p)$$

(this coincides with independent construction of Ppps w/ intensities $\lambda p$, $\lambda(1-p)$ up to terms of size $O(1/n^2)$).

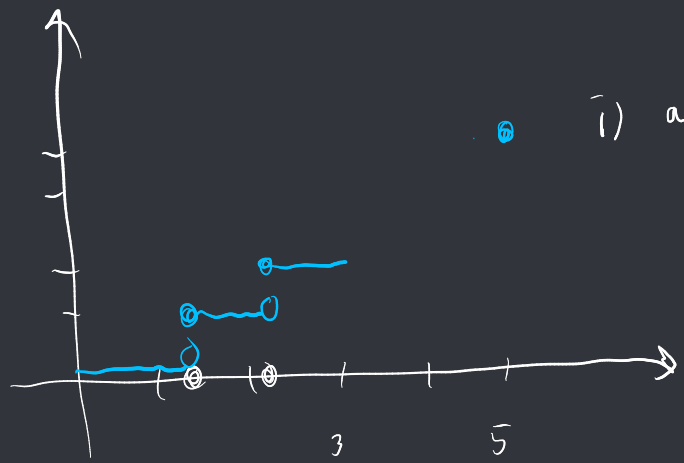
examples: let $X(t)$ = # points in $[0,t]$ Poisson process w/ intensity $\lambda$.

find i) $P(X(5)=4 \mid X(3)=2)$

ii) $P(X(3) > X(2))$

iii) $P(X(5) - X(4) = 3 \mid X(2) - X(1) = 4)$

i)




i) asks for

$P(2 \text{ points in } [3,5])$

$= P(\text{Poiss}(\lambda(5-3)) = 2)$

$= P(\text{Poiss}(2\lambda) = 2)$

$= e^{-2\lambda} (2\lambda)^2 / 2!$

ii) asks for $P(\geq 1 \text{ point in } [2,3])$

$$= 1 - P(\text{Poiss}(\lambda) = 0)$$

$$= 1 - e^{-\lambda}$$

iii): $P(\#\text{points in } [4,5] = 3 \mid \#\text{points in } [1,2] = 4)$

$$= P(\#\text{points in } [4,5] = 3)$$

$$= P(\text{Poiss}(\lambda) = 3) = e^{-\lambda} \lambda^3 / 3!$$

Consider process in which cars go by @ rate 3/min.
& trucks go by @ rate 2/min independently.

i) what is i) P( first vehicle is truck)

ii) P( third vehicle car | first vehicle is truck)

iii) P( 2 cars in [1,3] | 4 trucks in [1,3])

iv) P( no vehicles in first 2 min).

(i) $P(\min(W_1, W_2) = W_1)$

(where $W_1 \overset{d}{=} \exp(2)$ $W_2 \overset{d}{=} \exp(3)$

$= \frac{2}{2+3} = 2/5$

(ii)

let $V_k$ = $k^{th}$ vehicle (car vs truck)

What is distribution of $(V_k)$?

Distribution of blue/red points can be obtained by thinning Ppp w/ intensity 5 by labelling each point blue (car) wp. $3/5$.

In particular, the labels are i.i.d. w/ $P(\text{blue}) = 3/5$.

so $P(3^{rd} \text{ is car} \mid 1^{st} \text{ is truck})$

$= P(3^{rd} \text{ is car}) = 3/5$.

(iii) $= P(2 \text{ cars in } [1,3])$ (processes are independent)

$= P(\text{Poiss}(3 \cdot (3-1)) = 2)$

$= e^{-6} \cdot 6^2 / 2!$

(iv) $= P(\text{Poiss}(5 \cdot 2) = 0) = e^{-10}$

2+3 total intensity

length of time interval.

Viewing points as stopping times

Let $(w_i)$ i.i.d. $\exp(\lambda)$ & $t_n = \sum_{i=1}^{n} w_i$

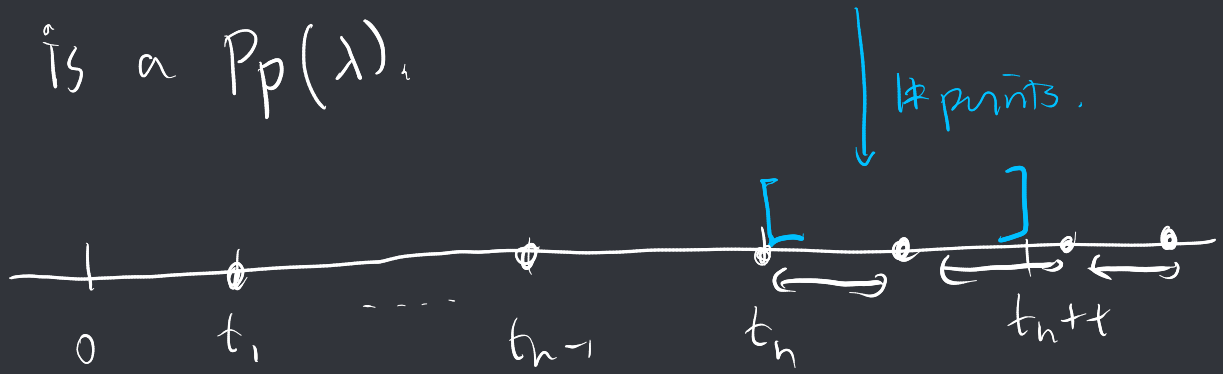
so that $(t_n)$ is a $PPP(\lambda)$

and $X(t) := \left| \{t_n\}_{n=1}^{\infty} \cap [0,t] \right|$ is a

Poisson process w/ intensity $\lambda$,

then for each $n$, $\left( \underbrace{X(t_n + t) - X(t_n)} \right)_{t \in \mathbb{R}}$

is a $Pp(\lambda)$.




reason: $t_{n+1} - t_n,\ t_{n+2} - t_{n+1},\ \ldots$

are i.i.d. $\exp(\lambda)$.

ex: in cars/trucks model,

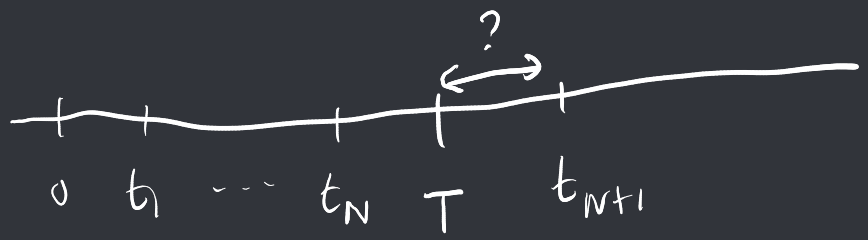
$$\left(\begin{array}{c}\#\text{ cars in first 2 min} \\ \text{after } 3^{\text{rd}} \text{ car}\end{array}\right) \overset{d}{=} \left(\begin{array}{c}\#\text{ cars in} \\ \text{first 2 min.}\end{array}\right)$$

also true if replace "3rd car" with "3rd truck"

(exercise: what's different?)

* <u>Exercise</u>: from $(W_i)$ & $(t_n)$ construction of $PPP(\lambda)$, show that $\forall T > 0$,

$\#\text{points in } [T, T+t] \overset{d}{=} \text{Poisson}(\lambda t)$




$N = \max\{n : t_n < T\}$

argue that $\overset{?}{\longleftrightarrow}$ is $\exp(\lambda)$.

(memoryless property)

# Compound Poisson process

to a Ppp attach to each point an independent random variable from an i.i.d. sequence:

Have $(t_n)$, $(u_n)$

$\uparrow$ Ppp$(\lambda)$

$\uparrow$ i.i.d. sequence.

Interested in

$$R(t) := \sum_{n=1}^{X(t)} u_n \longleftarrow \text{counting process}$$

$$\left( X(t) = \max\{n : t_n \le t\} \right)$$

<u>ex</u>: customers arrive @ rate $\lambda$

each spends random amount $u_n$.

<u>Q</u>: what is mean & variance of $R(t)$ for fixed $t$?

Suppose $N$, $(u_n)$ are independent. (i.i.d.)

Compute

$$E\left[\sum_{n=1}^{N} u_n\right], \quad Var\left[\sum_{n=1}^{N} u_n\right]$$

if $N$ deterministic,

$$= N E[u_n] \qquad = N Var(u_n)$$

if $N$ random, mean same but variance larger.

<u>Mean</u>: condition on value of $N$

$$E\left[\sum_{i=1}^{N} u_i\right] = \sum_n E\left[\sum_{i=1}^{n} u_i \mid N=n\right] P(N=n)$$

$$= \sum_n n\, E[u_i]\, P(N=n)$$

$$= E[u_i] \sum_n n\, P(N=n)$$

$$= E[u_i]\, E[N]$$

<u>Variance</u>: compute second moment

(Answer will be

$$E[N]\,Var(u_i) + Var(N)\,E[u_i^2]$$ )

$$E\left[\left(\sum_{i=1}^{N} u_i\right)^2\right] = E\left[\left(\sum_{i=1}^{n} u_i\right)^2 \middle| N=n\right] P(N=n)$$

now

$$\left(\sum_{i=1}^{n} u_i\right)^2 = \overbrace{u_1^2 + \cdots + u_n^2}^{= n\,E[u_i^2]}$$

$$+ \underbrace{\sum_{i \neq j} u_i u_j}$$

# terms $= n(n-1)$

each has $E[u_i u_j]$

$= E[u_i]E[u_j]$